

FitBridge v2 – Complete Technical PRD & Roadmap

1. Executive Summary

FitBridge v2 evolves into a scalable, adaptive AI fitness intelligence platform capable of serving 100K–1M users. The system adopts hybrid AI architecture, adaptive state engines, microservices separation, and observability-first engineering.

2. Product Goals

- Hybrid AI pipeline (Rule Engine + Template Engine + LLM narrative layer)
- Adaptive personalization state machine
- Scalable microservices architecture
- Cost-efficient inference layer
- Observability & performance monitoring
- Secure storage of health data

3. System Architecture Overview

- Frontend: React / React Native with streaming responses
- API Gateway: JWT validation, rate limiting, request validation
- Microservices: Auth, Profile, Plan Engine, Adaptive Engine, Inference, Analytics
- Database: PostgreSQL for structured data
- Caching: Redis for plan reuse
- Inference Service: Isolated LLM layer with cost tracking and fallback

4. Core Functional Modules

- User Profiling Engine with validation rules
- Derived Metrics: BMI, Progress Score, Consistency Score, Fatigue Index
- Adaptive State Machine: Beginner → Consistent → Plateau → Fatigued → Advanced
- Hybrid Plan Engine: Rule Filter → Template Selector → Adaptive Modifier → LLM Formatting

5. API Contracts

- POST /auth/signup
- POST /auth/login
- POST /plan/workout
- POST /plan/diet
- POST /activity/log
- GET /adaptive/state/{user_id}
- POST /analytics/event

6. Database Schema Summary

- users
- profiles
- user_metrics
- workout_plans
- diet_plans
- activity_logs
- llm_usage_logs

7. Non-Functional Requirements

- Uptime: 99.9%
- Plan Generation Latency: <2.5 seconds
- Backend API Latency: <200ms
- Cost per Active User: < █15/month target
- Horizontal Scalability: 1M users
- Encrypted storage of health data

8. Implementation Roadmap

- Phase 1 (Month 1): Input validation, hybrid logic, monitoring setup
- Phase 2 (Month 2–3): Adaptive engine, caching layer, cost tracking

- Phase 3 (Month 4+): Autoscaling, A/B testing, premium inference tier

9. Success Metrics (KPIs)

- $\geq 70\%$ Plan Acceptance Rate
- $\geq 60\%$ Week-2 Retention
- < 2.5 sec average plan generation time
- $< 1\%$ hallucination incidents
- LLM cost within monthly target budget

10. Conclusion

FitBridge v2 transitions from a simple AI workout generator into a scalable, adaptive AI fitness intelligence platform with production-grade architecture, cost awareness, and personalization depth.